Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – VI

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Part A - Program

a. Create a simple website for a College Information Portal and convert it into a Progressive Web Application.

Code:

manifest.json:

```
{
  "name": "HTML TO PWA",
  "short_name": "HTP",
  "start_url": "/",
  "display": "standalone",
  "icons": [
    {
      "src": "/icons/icon1.png",
      "sizes": "128x128",
      "type": "image/png",
      "purpose": "any"
    },
    {
      "src": "/icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png",
      "purpose": "any"
    }
  ]
}
```

service-worker.js:

```
self.addEventListener("install", e => {
   console.log("Service Worker installed");
});

self.addEventListener("fetch", e => {
   e.respondWith(fetch(e.request));
});
```

_Layout.cshtml:

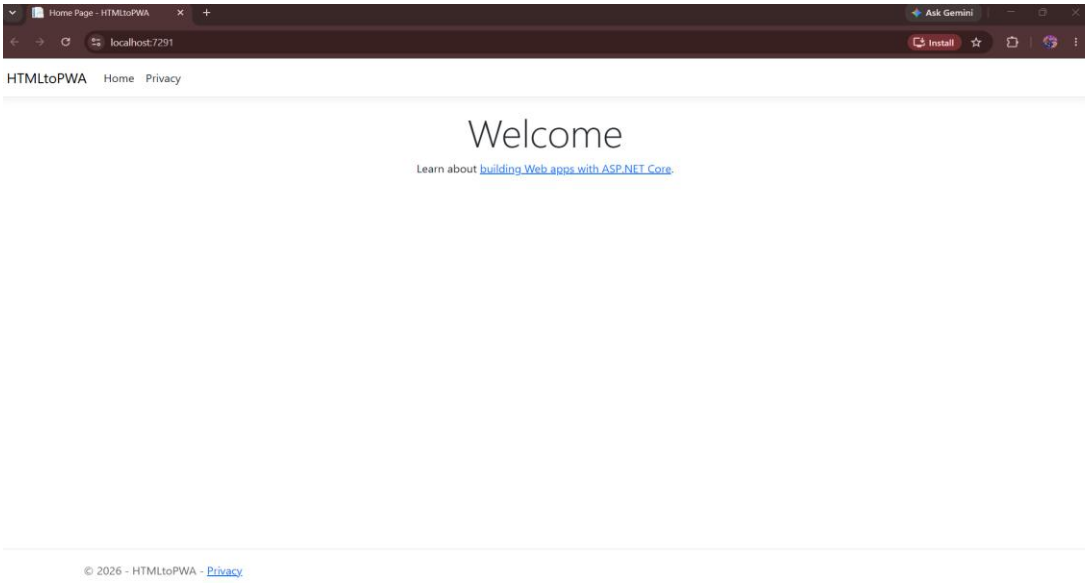
Inside the <head> **section:**

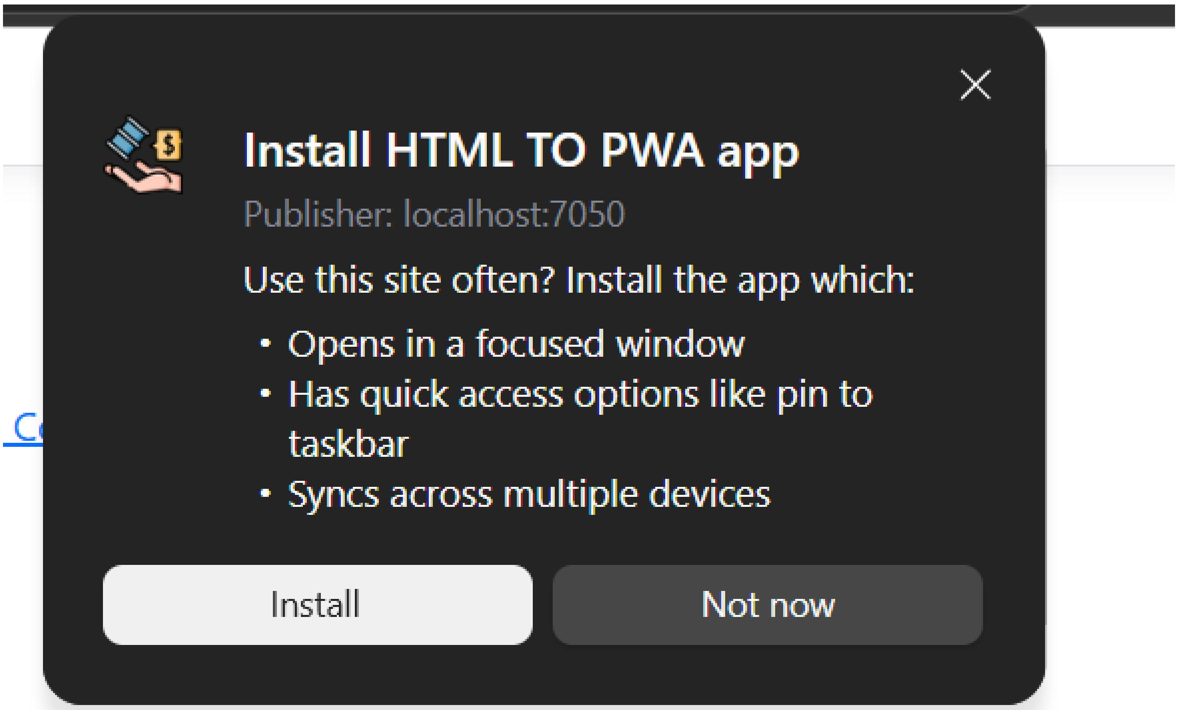
```
<link rel="manifest" href="/manifest.json" />
```

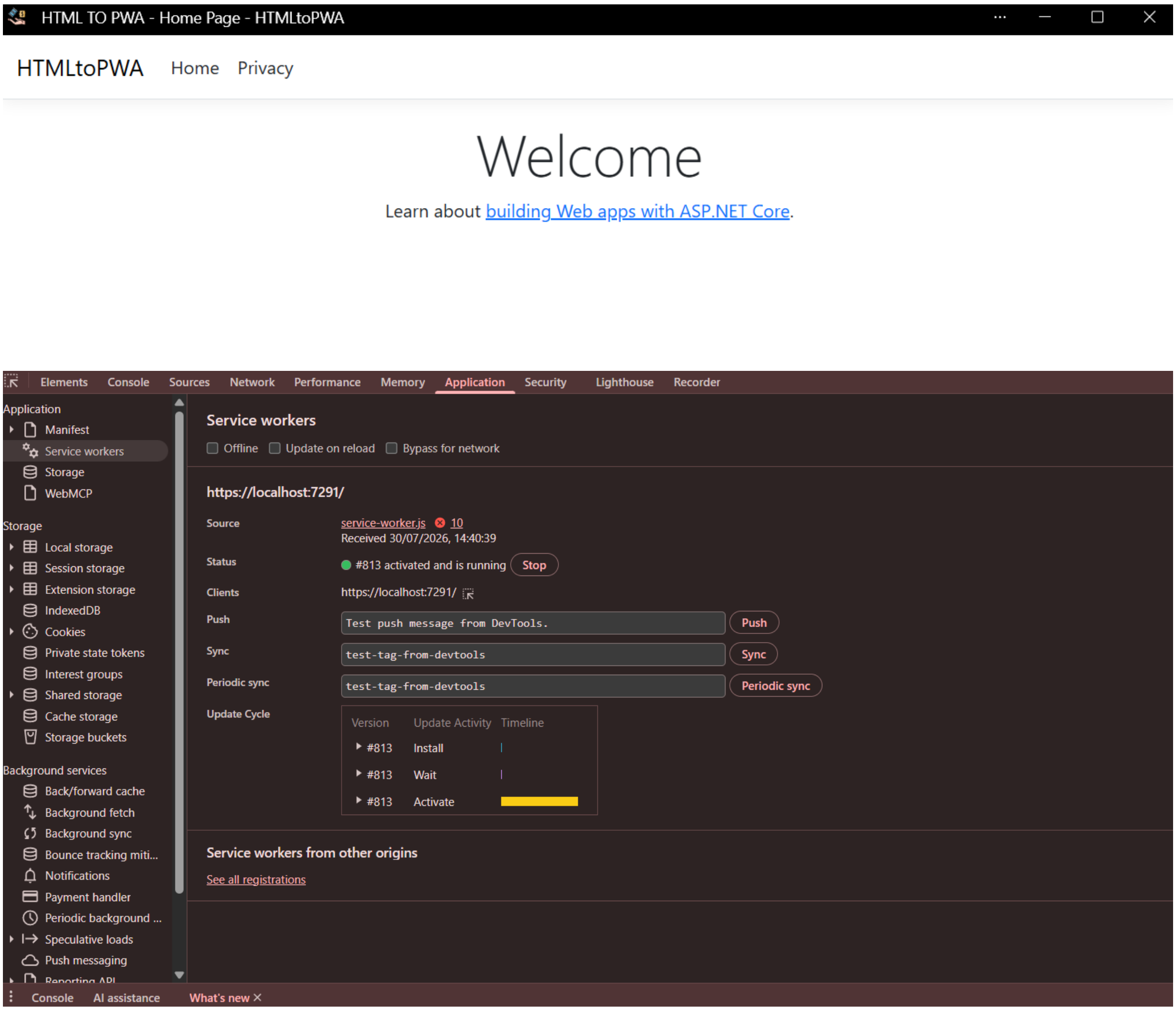
Before the closing </body> **tag:**

```
<script>
    if('serviceWorker' in navigator){
        navigator.serviceWorker.register('/service-worker.js');
    }
 </script>
```

Output:






HTML TO PWA - Home Page - HTMLtoPWA
HTMLtoPWA
Home
Privacy
Welcome
Learn about building Web apps with ASP.NET Core.
Elements
Console
Sources
Network
Performance
Memory
Application
Security
Lighthouse
Recorder
Application
Manifest
Service workers
Storage
WebMCP
Storage
Local storage
Session storage
Extension storage
IndexedDB
Cookies
Private state tokens
Interest groups
Shared storage
Cache storage
Storage buckets
Background services
Back/forward cache
Background fetch
Background sync
Bounce tracking miti...
Notifications
Payment handler
Periodic background ...
Speculative loads
Push messaging
Service workers
Offline
Update on reload
Bypass for network
https://localhost:7291/
Source
service-worker.js
10
Received 30/07/2026, 14:40:39
Status
#813 activated and is running
Stop
Clients
https://localhost:7291/
Push
Test push message from DevTools.
Push
Sync
test-tag-from-devtools
Sync
Periodic sync
test-tag-from-devtools
Periodic sync
Update Cycle
Version
Update Activity
Timeline
#813
Install
#813
Wait
#813
Activate
Service workers from other origins
See all registrations
Console
AI assistance
What's new

b. Create a PWA that displays a custom offline page when the user is not connected to the internet.

Code:

offline.html :

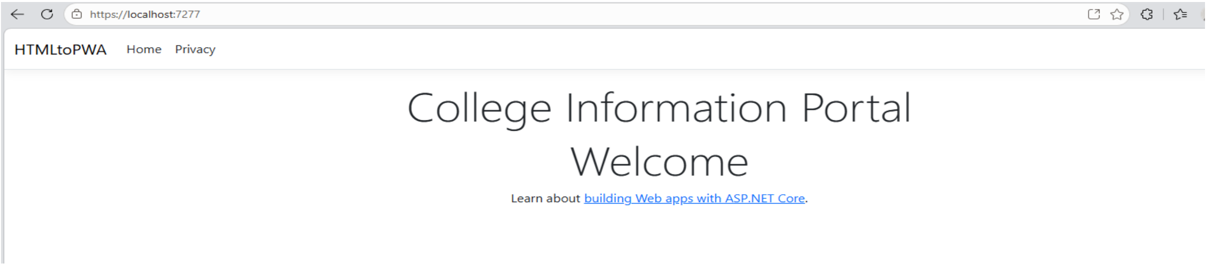
```
<!DOCTYPE html>
<html>
<head>
  <title>Offline</title>
</head>
<body style="text-align:center; margin-top:100px; font-family:Arial;">
  <h1>You are Offline</h1>
</body>
</html>
```

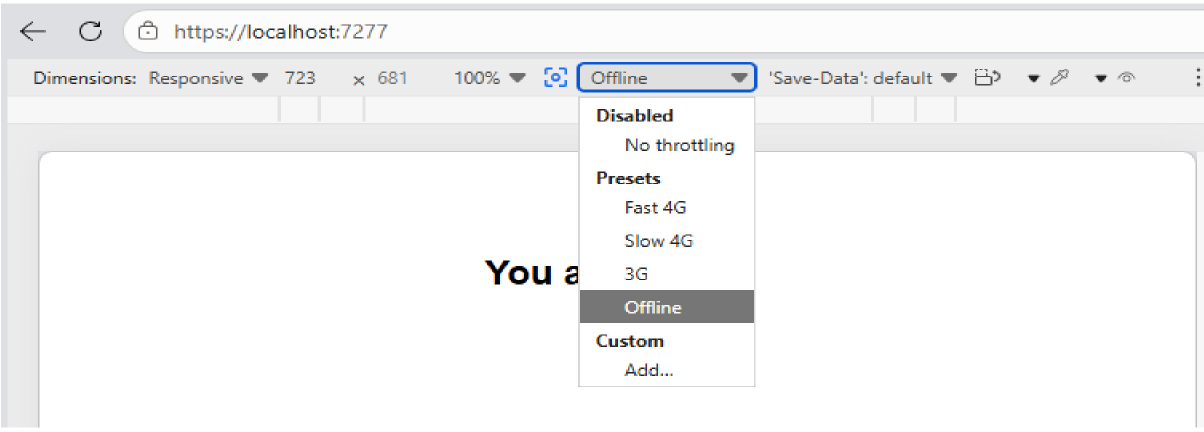
service-worker.js :

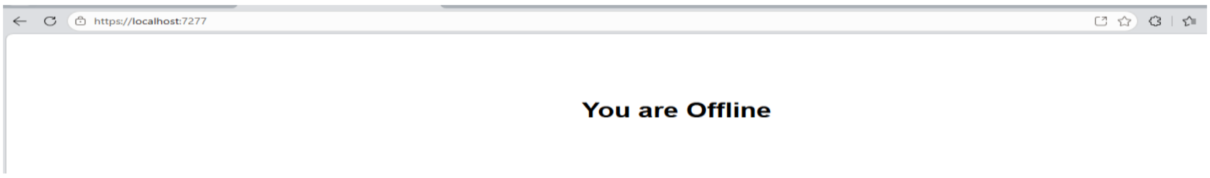
```
const CACHE = "v1'";
self.addEventListener("install", event => {
  event.waitUntil(
    caches.open("offline-cache").then(cache => {
      return cache.addAll([
        "/",
        "/offline.html"
      ]);
    }));

  console.log("Service workers Installed ")
});
self.addEventListener("fetch", event => {
  event.respondWith(
    fetch(event.request).catch(() => {
      return caches.match("/offline.html");
    })
  );
});
```

Output:


https://localhost:7277
HTMLtoPWA
Home
Privacy
College Information Portal
Welcome
Learn about building Web apps with ASP.NET Core.


https://localhost:7277
Dimensions: Responsive
723
x 681
100%
Offline
'Save-Data': default
Disabled
No throttling
Presets
Fast 4G
Slow 4G
3G
Offline
Custom
Add...
You a


https://localhost:7277
You are Offline

Part C – Curiosity Question

Your college has a website that students use every day to check notices, timetables, and study materials.

Many students say:

"It would be easier if this website worked like a mobile app."

The college does not want to spend money on developing a separate Android or iPhone app.

Task

How can you convert the college website into an app-like experience that students can install on their phones?

Hint: Use a **Progressive Web Application (PWA)**.